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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations

December 2025

MA2001-1 – STATISTICS AND PROBABILITY THEORY
(For CS/IS/AM/CC/AD/CB/RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. The pdf of a random variable is $2x, 0 < x < 1$ then $P(0 < X < 1)$ is _____
A) -1
B) 0
C) $\frac{1}{2}$
D) 1
2. Which of the following are not examples of a discrete random variable?
A) Number of friends you have
B) the number of cars you own
C) How tall you are?
D) the number of people in your class today
3. If X and Y are independent random variable then $Cov(X, Y) =$
A) $E(X+Y)$
B) 0
C) $E(X-Y)$
D) $E(X)E(Y)$
4. If $V[X] = 2$ then $V[2X + 1]$ is
A) 8
B) 4
C) 5
D) 9
5. The standard deviation of binomial distribution is
A) npq
B) np
C) $\sqrt{npq}$
D) 1
6. For standard normal distribution mean is
A) 0.5
B) 0
C) -1
D) 1
7. The mean of an exponential distribution is 2 then standard deviation is
A) $1/2$
B) 2
C) 4
D) $1/4$
8. If p is a probability of success and q is a probability of failure then $(p + q)^5$ is
A) 0
B) 1
C) 2^n
D) 2
9. When sample size $n < 30$, we use
A) normal
B) Student t-distribution
C) Chi-square distribution
D) Exponential
10. Samples of size 25 are selected from a population with mean 40 and standard deviation 7.5. The mean of the sampling distribution of sample means is
A) 7.5
B) 8
C) 25
D) 40
11. The degrees of freedom for $\frac{ns^2}{\sigma^2}$ which is in chi-square distribution is
A) $n-1$
B) n^2
C) 1
D) 0
12. A statistic is
A) a function of population units
B) a characteristic of a population
C) a function of sample observations
D) a part of a population



13. If the regression lines are $x = 0.7y + 5.2$ and $y = 0.3x + 2.8$ then the $(\bar{x}, \bar{y})$ is
 A) (1,1) B) (5.2,2.8)
 C) (0.7, 0.3) D) (9, 5.52)
14. The method used to fit a parabola to the given data is known as
 A) least square B) quadratic
 C) least cube D) bi quadratic
15. One among the normal equation for fitting of a straight line $y = a + bx$ is $\sum y =$
 A) $na + b \sum x^2$ B) $na + b \sum x$
 C) $n^2a + b \sum x$ D) $a + b \sum x$
16. The standard deviation of x and y are 6 and 8 respectively. The correlation coefficient between them is 0.6. Then the value of regression coefficient b_{yx} is
 A) 0.6 B) 0.8
 C) 1.5 D) 0.75
17. If the regression coefficient of x on y is $-\frac{1}{6}$, of Y on X is $-\frac{3}{2}$ then correlation coefficient is
 A) -0.5 B) 0.5
 C) 1 D) 0
18. Which of the following is a Stochastic matrix?
 A) $\begin{bmatrix} \frac{1}{3} & \frac{1}{5} & \frac{1}{4} \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{3} \end{bmatrix}$ B) $\begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$
 C) $\begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{3} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{bmatrix}$ D) $\begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{bmatrix}$
19. Which among the following vectors is a probability vector?
 A) $(1/12, 1/2, 1/6, 0, 1/4)$ B) $(1/2, 0, -2/3, 1/5, 3/5)$
 C) $(5/2, 0, 8/3, 1/6, 1/6)$ D) $(0.25, -0.25, 0.5, 0.5)$
20. Which of the following best describes queuing theory?
 A) The evaluation of service time costs B) The study of waiting lines
 C) The study of service times D) The study of arrival rates

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

Marks BT* CO* PO*

1. a) The probability distribution function of a random variable X is given by the following table

X	1	2	3	4	5	6	7
P(X)	3k	k	3k	5k	k	2k	3k

- i. find the value of k
 ii. find $P(1 \leq X \leq 5)$ and $E(2X - 1)$
- b) Obtain the mean and variance of the Poisson distribution.
- c) The errors in a certain length measuring device are known to be normally distributed with expected value zero and standard deviation 1 inch. What is the probability that the error in measurement will be
 i. greater than 1 inch
 ii. less than 3 inches?
2. a) Two continuous random variables X and Y have the following joint distribution
 $f(x, y) = \begin{cases} kxy & 0 \leq x \leq 4, 1 \leq y \leq 5 \\ 0 & \text{otherwise} \end{cases}$
 i. find 'k' ii. $P(x \geq 2, y \leq 3)$

- b) A box contains 12 items of which 4 are defective. A sample of 3 items are selected from the box. Let X denote the number of defective items in the sample. Find the probability distribution of X . Also determine the mean and the standard deviation of the distribution.
- c) Prove that if $M_X(t)$ is the moment generating function of a random variable X , then $M'_X(0)$ is $E(X)$ and $M''_X(0)$ is $E(X^2)$.

6 L3 1 2
4 L2 2 2

3. a) Suppose that the following table represents the joint p.d.f.

X \ Y	1	2	3
1	1/8	1/8	2/8
2	3/8	0	0
3	0	1/8	0

- i. Evaluate the marginal distribution of X and Y
- ii. $E(X)$
- iii. $P(X=2|Y=1)$
- iv. check for in dependency.
- b) The probability that a pen manufactured by a company will be defective is $\frac{1}{10}$. If 12 such pens are manufactured, find the probability that:
- i. at most two will be defective
- ii. at least three will be defective
- iii. none will be defective.
- c) The daily amount X of coffee, in litres dispensed by a machine is uniformly distributed with $a = 7$, $b = 10$. Determine the probability that the amount of coffee dispensed by the machine will be
- i. at most 8.8
- ii. more than 7.4 but less than 9.5
- iii. at least 8.5 litres.

6 L2 1 2
6 L3 2 1
4 L2 2 2

Unit – II

4. a) Calculate the Karl Pearsons coefficient of correlation for the following data

x	1	2	3	4	5	6	7
y	9	8	10	12	11	13	14

- b) Fit a straight line $y=a+bx$ to the following data by the method of least squares:

x	12	15	21	25
y	50	70	100	120

- c) If $\bar{X}$ is the mean of a random sample of size n from a population with mean μ and variance σ^2 , then prove that $E(\bar{X}) = \mu$ and $V(\bar{X}) = \frac{\sigma^2}{n}$

6 L2 4 2
6 L3 4 1
4 L2 3 2

5. a) i. State Central Limit Theorem.
- ii. If $\bar{X}$ is the mean of a random sample of size n from a normal distribution with mean μ and variance 100. Find n so that $P[\mu - 5 < \bar{X} < \mu + 5] = 0.954$.
- b) For the regression lines $4x-5y+33=0$, $20x-9y=107$ find the mean values of x and y , coefficient of correlation between x and y .
- c) A random sample of size 20 from a normal distribution $N(\mu, \sigma^2)$ gives $\bar{X}=3.2$, $S^2 = 4.24$. Determine a 95% confidence interval for σ^2 .

6 L2 3 2
6 L3 4 1
4 L2 3 2

6. a) Fit a parabola $y = ax^2 + bx + c$ for the following data by the method of least squares.

x	0	1	2	3	4
y	1	1.8	1.3	2.5	2.3

6 L2 4 2

- b) Compute the rank correlation coefficient for the following data:

x	68	64	75	50	64	80	75	40	55	64
y	62	58	68	45	81	60	68	48	50	70

6 L2 4 1

- c) The random variable X has t-distribution with 14 degrees of freedom. Find the probability of
- $|X| \geq 1.34$
- .

4 L3 3 2

Unit – III

7. a) A software engineer goes to his workplace every day either by bike or by car. He never goes by bike on two consecutive days; but if he goes by car on a day then he is equally likely to go by car or bike on the next day. Find the transition matrix for the chain of the mode of transport he uses. If car is used on the first day of the week, find the probability that after four days

i. bike is used ii. car is used.

6 L2 5 2

- b) Find the unique fixed probability vector for the regular stochastic matrix

$$P = \begin{bmatrix} 0 & 2/3 & 1/3 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{bmatrix}$$

6 L2 5 1

- c) Consider the Markov Chain with 3 states
- $S=\{1,2,3\}$
- that has the following transition matrix

$$P = \begin{bmatrix} 1/2 & 1/4 & 1/4 \\ 1/3 & 0 & 2/3 \\ 1/2 & 1/2 & 0 \end{bmatrix}$$

i. Draw the state transition diagram for this chain.

ii. If we know $P(X_1 = 1) = P(X_1 = 2) = \frac{1}{4}$ find $P(X_1 = 3, X_2 = 2, X_3 = 1)$

4 L3 5 2

8. a) Assume that the computer system is in one of the three states: busy, idle or undergoing repair denoted by states 0,1, 2. Observing its state at a certain specific time on each day, it is found that the system approximately behaves

like a Markov chain with the transition probability matrix $A = \begin{bmatrix} 0.6 & 0.2 & 0.2 \\ 0.1 & 0.8 & 0.1 \\ 0.6 & 0.0 & 0.4 \end{bmatrix}$

Determine the steady state probabilities.

6 L2 5 2

- b) Given transition probability matrix of a Markov Chain
- $P = \begin{bmatrix} 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \\ 0.3 & 0.4 & 0.3 \end{bmatrix}$

With initial probability distribution $p^0 = (0.7, 0.2, 0.1)$.

Find

i. $P(X_2 = 3)$ ii. $P(X_1 = 3)$ iii. $P(X_3 = 2, X_2 = 3, X_1 = 3, X_0 = 2)$.

6 L2 5 1

- c) i. Define a regular stochastic matrix.

ii. If $A = \begin{bmatrix} a_1 & a_2 \\ b_1 & b_2 \end{bmatrix}$ is a stochastic matrix and $V = [v_1 \ v_2]$ is a probability vector, show that VA is also a probability vector.

4 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
